AZURE DEVOPS KICKOFF

BHARATH

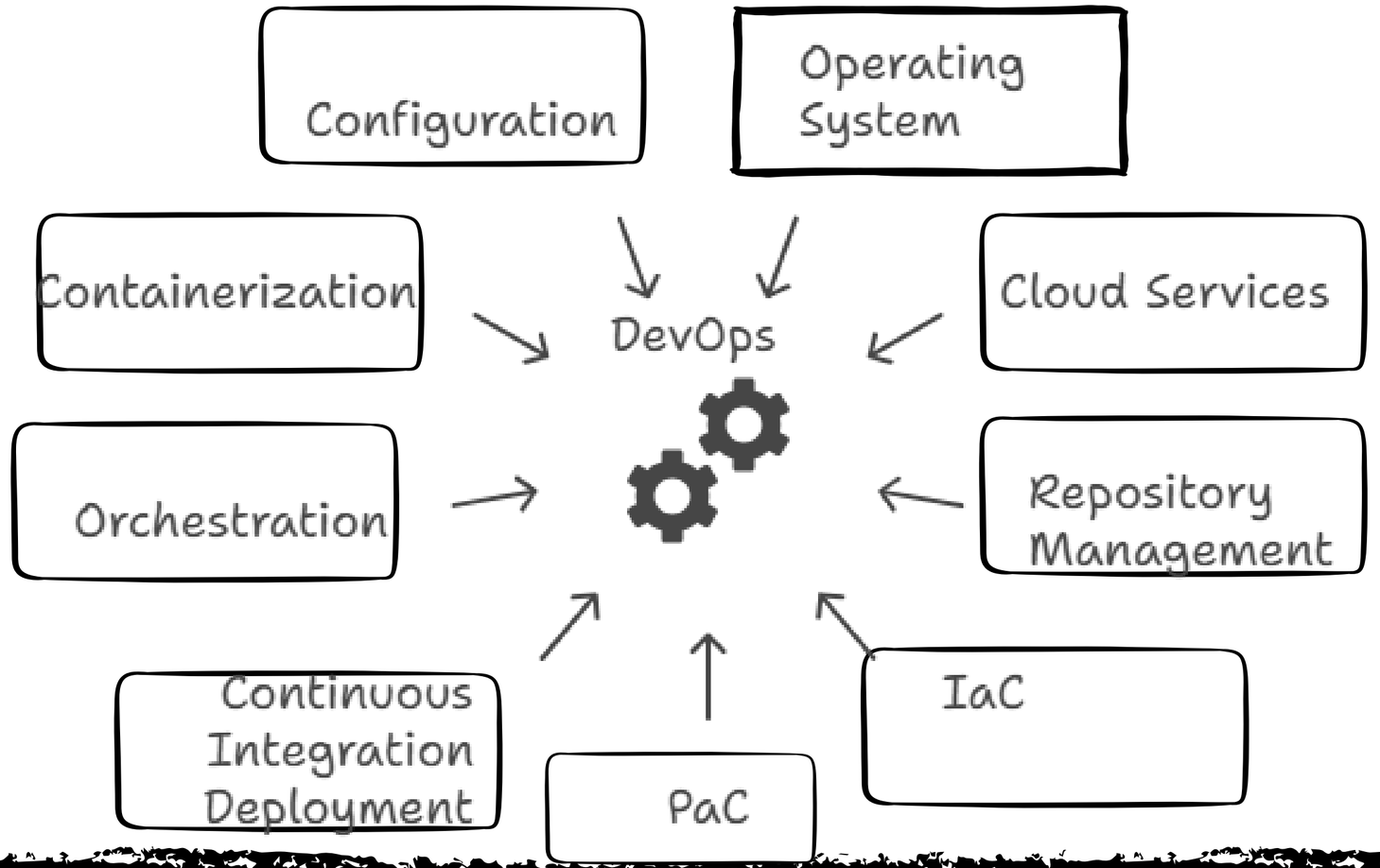
ABOUT ME

- 14+ YEARS OF INDUSTRY EXPERIENCE IN IT INFRASTRUCTURE & CLOUD
- 10+ YEARS OF TEACHING: VMWARE, AZURE,AWS CLOUD, AZURE DEVOPS
- TRAINED OVER 6,00+ STUDENTS GLOBALLY
- PASSIONATE ABOUT SIMPLIFYING COMPLEX TECH

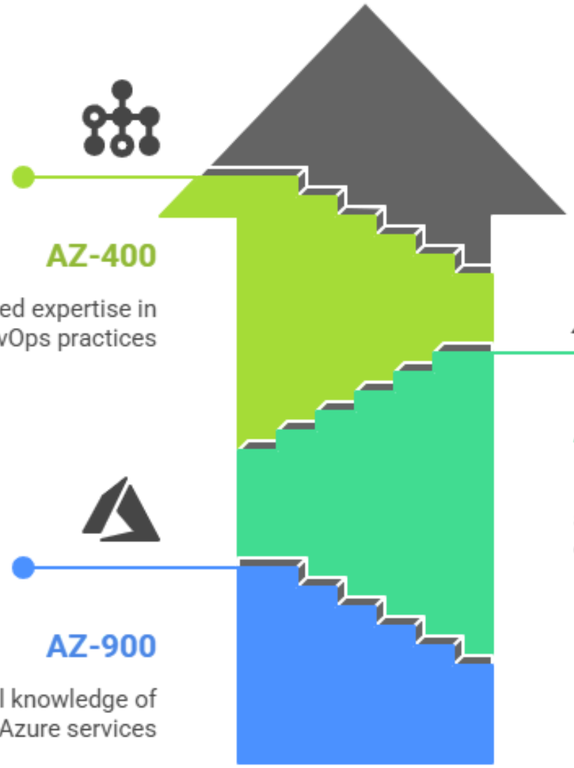
LinkedIn/BHARATH-REDDY329


WHAT THE HELL IS DEVOPS

- INTRODUCTION TO CLOUD COMPUTING
- MICROSOFT AZURE FUNDAMENTALS & ADMIN
- LINUX
- POWERSHELL
- AZURE CLI
- GIT AND GITHUB/AZURE REPOS
- INTRODUCTION TO DEVOPS
- AZURE DEVOPS/GITHUB ACTIONS
- DOCKER FOR DEVOPS
- KUBERNETES FOR DEVOPS
- MAVEN
- AZURE PIPELINES
- TERRAFORM
- AZURE MONITOR


Configuration
Operating System
Containerization
DevOps
Cloud Services
Orchestration
Repository Management
Continuous Integration Deployment
PaC
IaC

Azure DevOps Certification Journey



AZ-400

Advanced expertise in DevOps practices

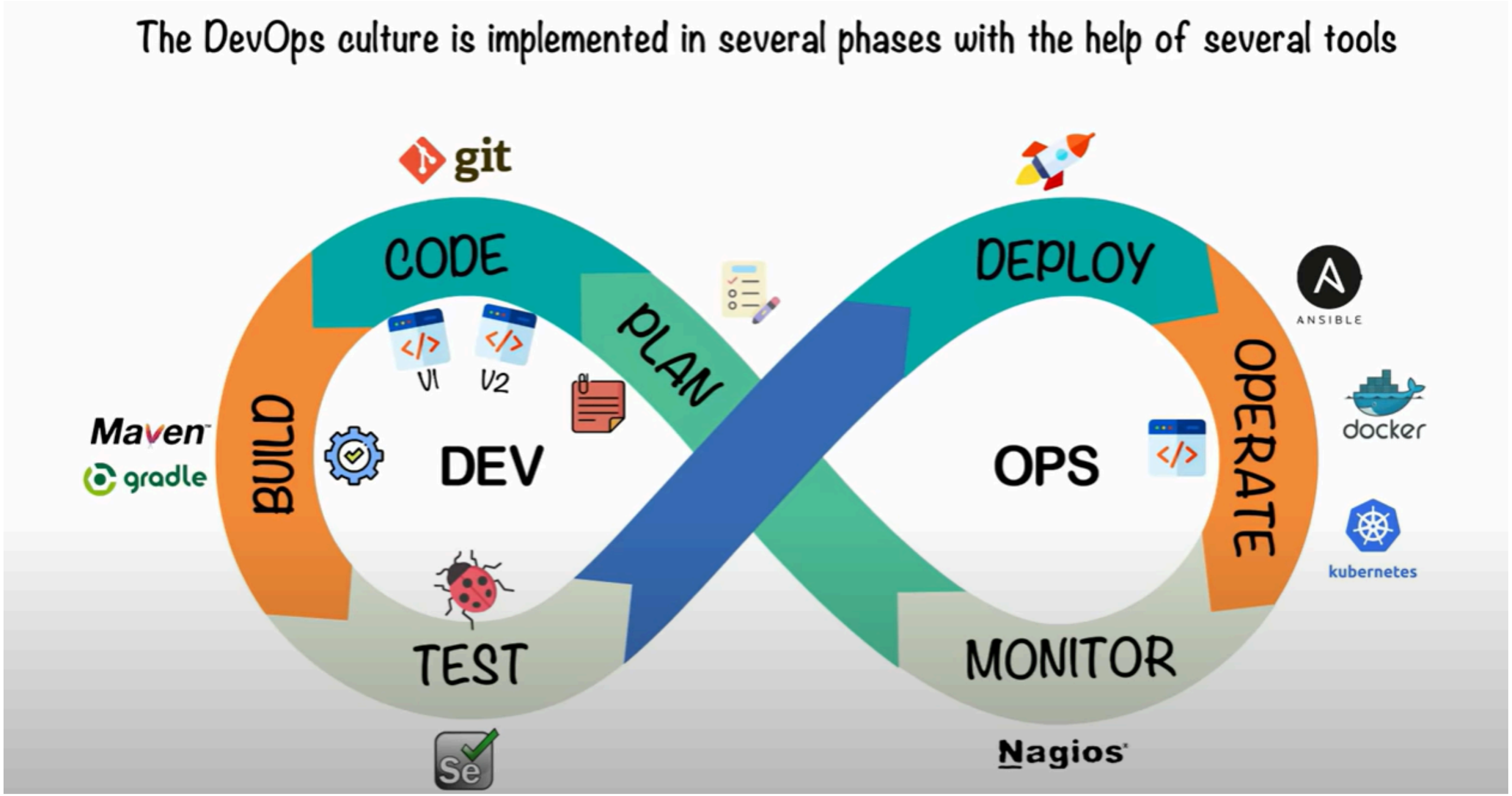
AZ-104/204

Intermediate skills in Azure administration or development

AZ-900

Foundational knowledge of Azure services

Made with Napkin


The DevOps culture is implemented in several phases with the help of several tools
git
CODE
PLAN
BUILD
V1
V2
DEV
Maven
gradle
TEST
Se
DEPLOY
OPERATE
OPS
MONITOR
Nagios
ANSIBLE
docker
kubernetes

DevOps Stage	Azure DevOps Tool	Replaces / Competes With
Version Control	Azure Repos	GitHub, Bitbucket, GitLab
CI/CD Pipelines	Azure Pipelines (YAML or Classic)	Jenkins, CircleCI, Travis CI
Work Tracking	Azure Boards	Jira, Trello
Artifact Storage	Azure Artifacts	JFrog, Nexus, GitHub Packages
Test Management	Azure Test Plans	Selenium, TestRail, QTest
IaC (Infra Mgmt)	Bicep, ARM Templates	Terraform, CloudFormation
Monitoring	Azure Monitor, Application Insights	Prometheus, Datadog, ELK, Grafana
Source Control (Alt)	GitHub Enterprise (Owned by Microsoft)	GitLab, Bitbucket

AZURE AI

AI 900

- **INTRODUCTION TO AI & ML**

  AI MAKES SYSTEMS SMART; ML HELPS THEM LEARN FROM DATA AUTOMATICALLY.

- **AZURE AI SERVICES**

  PREBUILT APIS AND TOOLS TO ADD INTELLIGENCE WITHOUT CODING ML FROM SCRATCH.

- **AZURE AI VISION**

  IMAGE & VIDEO ANALYSIS – DETECT OBJECTS, TEXT, FACES, SCENES.

- **AZURE AI FOUNDRY**

  END-TO-END PLATFORM TO BUILD, DEPLOY, AND MANAGE GENERATIVE & CUSTOM AI MODELS.

- **FACIAL RECOGNITION**

  IDENTIFY/VERIFY PEOPLE AND DETECT FACIAL ATTRIBUTES SECURELY.

- **GENERATIVE AI**

  CREATE CONTENT LIKE TEXT, CODE, IMAGES USING LARGE AI MODELS.

- **DEEP LEARNING**

  NEURAL NETWORKS THAT SOLVE COMPLEX TASKS LIKE VISION, SPEECH, AND NLP.

- **MACHINE LEARNING**

  TRAIN, DEPLOY, AND MONITOR ML MODELS USING AZURE MACHINE LEARNING & MLOPS.

DURATION

65 DAYS

- DURATION: 65 DAYS
- 85% PRACTICAL AND 15% THEORY
- REAL-LIFE SCENARIOS: YOU WILL SIMULATE ACTUAL PROJECT WORKFLOWS
- RESUME REVIEW + LINKEDIN PROFILE TIPS
- INTERVIEW PREPARATION
- CAREER GUIDANCE
- 1:1 INTERACTION

WHO CAN LEARN

- 👶 FRESHERS: START STRONG IN A FUTURE-PROOF IT PATH
- 👨‍💼 EXPERIENCED PROFESSIONALS: UPGRADE OR SWITCH TO DEVOPS ROLES
- 👩‍👧 CAREER BREAK? RE-ENTER IT WITH A STRONG, MODERN SKILLSET
- 🎯 IDEAL FOR SYSTEM ADMINS, DEVELOPERS, TESTERS, SUPPORT ENGINEERS, OR CLOUD LEARNERS

WHAT IS DIFFERENT

- 🔥 NARRATION WITH REAL LIFE EXAMPLES
- 🧠PRACTICAL-FIRST APPROACH, NOT JUST THEORY
- 📽️ CLASS RECORDINGS, NOTES, DOUBT SESSIONS
- ✅ REGULAR UPDATES BASED ON LIVE INDUSTRY TRENDS


ANY DOUBTS!!!!!!!


DAY 1


ONCE UPON A TIME

ON PREMISE



CLOUD COMPUTING

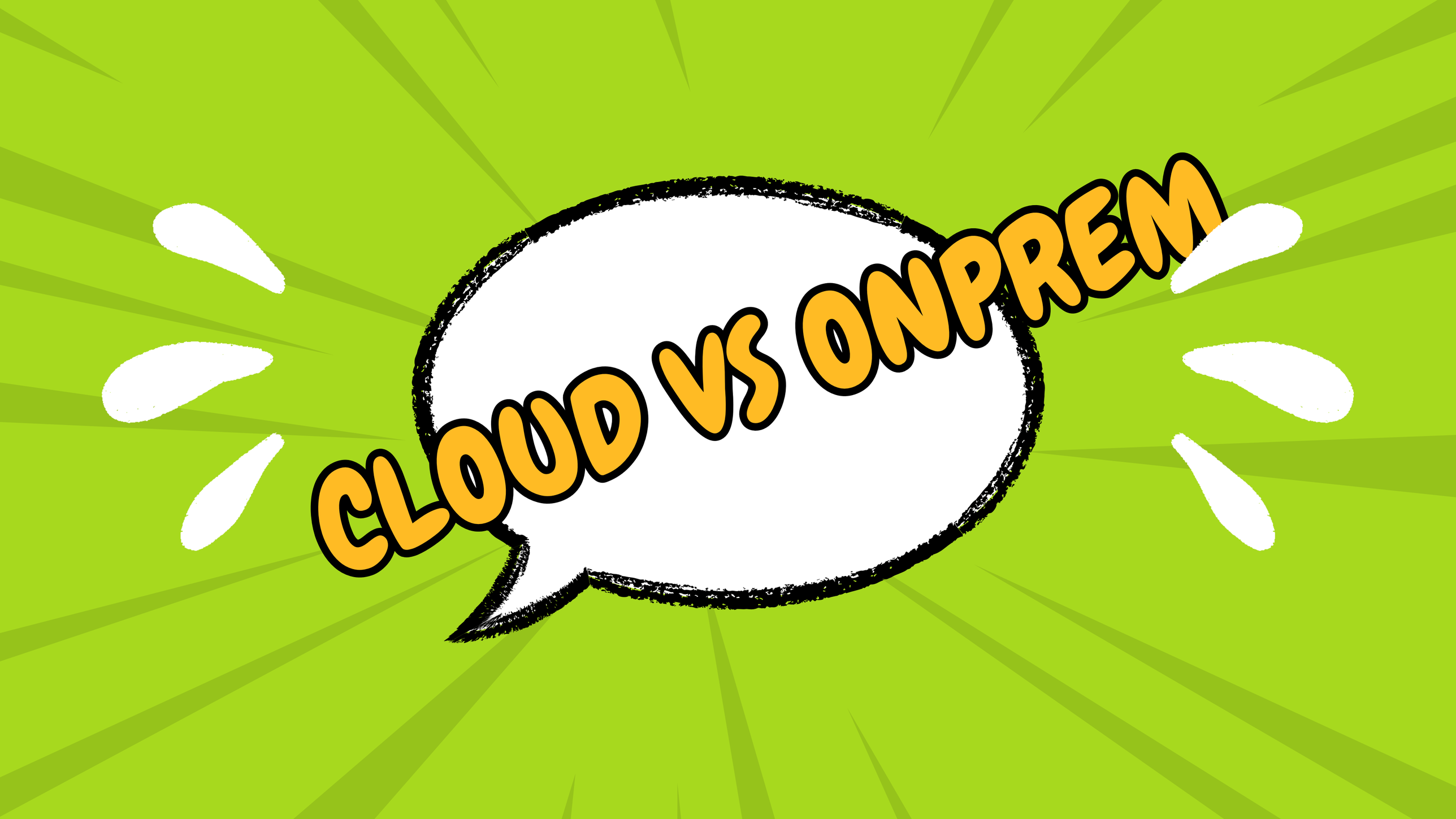
DEFINITION

- CLOUD COMPUTING IS THE DELIVERY OF COMPUTING SERVICES (LIKE SERVERS, STORAGE, DATABASES, NETWORKING, SOFTWARE) OVER THE INTERNET – "THE CLOUD" – ON-DEMAND.
- THINK OF CLOUD LIKE UBER – INSTEAD OF OWNING A CAR (SERVER), YOU REQUEST IT ONLY WHEN NEEDED AND PAY FOR WHAT YOU USE. NO MAINTENANCE, NO UPFRONT COST.

PROVIDERS

- AZURE
- AWS
- GCP
- SALESFORCE




CLOUD VS ONPREM

COMPARISION

On-Premises	Cloud
Requires physical space, servers	No physical infra required
High upfront investment	Pay-as-you-go model
Maintenance handled by you	Maintenance handled by cloud provider
Scaling is manual and costly	Scaling is instant and automated

OWNING A HOUSE (ON-PREMISES) VS. LIVING IN A SERVICE APARTMENT(CLOUD).

SERVICE APARTMENT PROVIDE AMENITIES, ROOM SERVICE, AND FLEXIBILITY – YOU JUST PAY FOR WHAT YOU USE.

CHARACTERSTICS

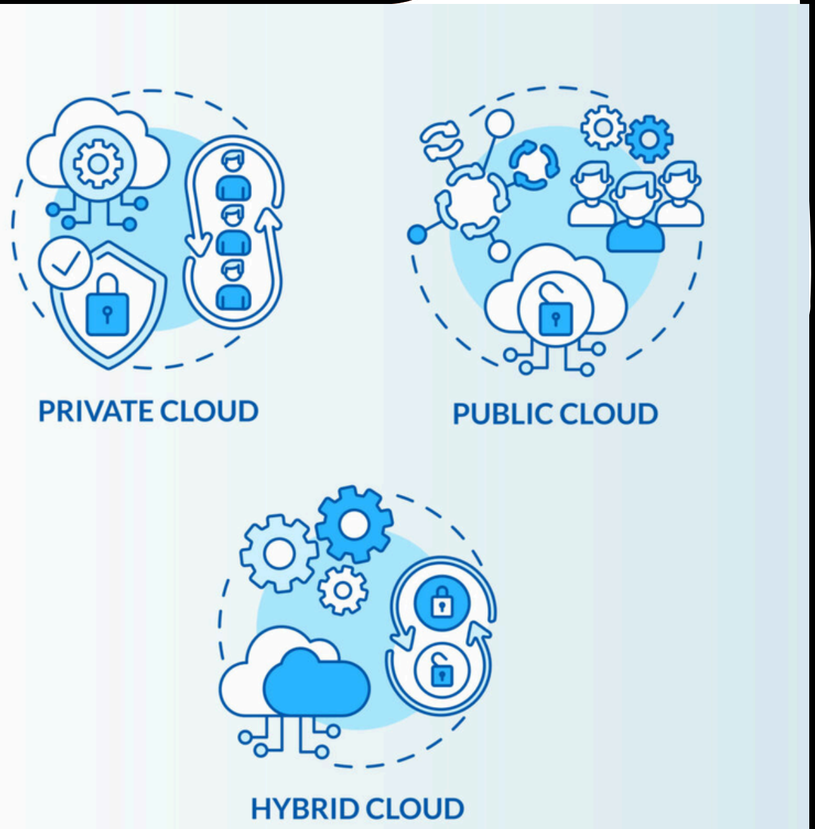
THE 5 PILLARS

- ON-DEMAND SELF-SERVICE – YOU CAN CREATE SERVERS ANYTIME
- BROAD NETWORK ACCESS – ACCESSIBLE FROM ANYWHERE
- RESOURCE POOLING – SHARED INFRA LIKE ELECTRICITY GRID
- RAPID ELASTICITY – AUTO SCALING UP/DOWN
- MEASURED SERVICE – YOU ONLY PAY FOR USAGE

MODELS OF CLOUD

PUBLIC, PRIVATE & HYBRID

- PUBLIC CLOUD: SHARED RESOURCES (E.G., AZURE, AWS)
- PRIVATE CLOUD: DEDICATED TO ONE ORGANIZATION (E.G., BANKS)
- HYBRID CLOUD: MIX OF PUBLIC + PRIVATE (MOST ENTERPRISES USE THIS)



SERVICES

IAAS, PAAS, SAAS

Model	What You Manage	What Vendor Manages	Example
IaaS	OS, apps, runtime	Servers, storage, network	Azure VM
PaaS	App & Data only	Everything else	Azure App Service
SaaS	Just use the software	All backend infra hidden	Outlook 365

- PUBLIC CLOUD: SHARED RESOURCES (E.G., AZURE, AWS)
- PRIVATE CLOUD: DEDICATED TO ONE ORGANIZATION (E.G., BANKS)
- HYBRID CLOUD: MIX OF PUBLIC + PRIVATE (MOST ENTERPRISES USE THIS)

BENEFITS

PUBLIC, PRIVATE & HYBRID

BENEFITS

- COST SAVINGS
- SCALABILITY
- ACCESSIBILITY
- SECURITY
- HIGH AVAILABILITY

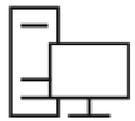
CONSIDERATIONS:

- DATA PRIVACY
- COMPLIANCE
- NETWORK DEPENDENCY

SLA's

VM SLA

99.9% with Premium Storage

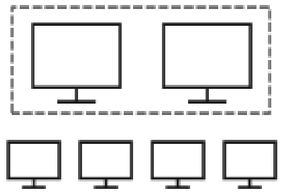


SINGLE VM

Easier lift and shift

VM SLA

99.95%

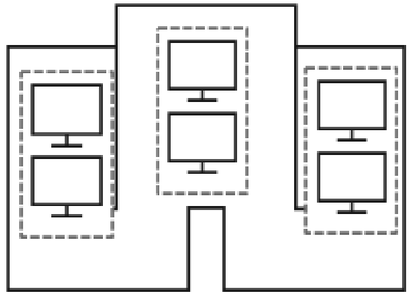


AVAILABILITY SETS

Protecting against failures within datacenters

VM SLA

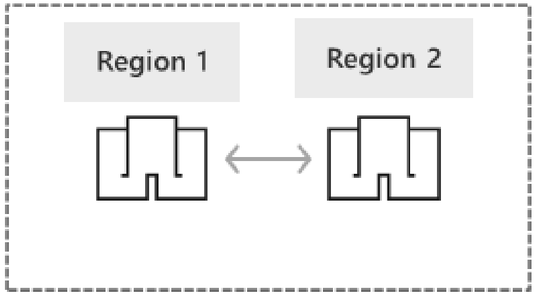
99.99%



AVAILABILITY ZONES

Protection from entire datacenter failures

MULTI-REGION DISASTER RECOVERY



REGION PAIRS

Regional protection within Data Residency Boundaries

TAKE AWAY

- “NOW THAT YOU KNOW THE WHAT, WHY, AND HOW OF CLOUD – FROM TOMORROW, WE’LL SEE HOW TO CREATE AN AZURE ACCOUNT AND PLAY WITH AZURE”

INTERVIEW QUESTIONS

- WHAT IS CLOUD COMPUTING, AND HOW IS IT DIFFERENT FROM TRADITIONAL ON-PREMISES INFRASTRUCTURE?
- EXPLAIN IAAS, PAAS, AND SAAS WITH REAL-TIME EXAMPLES. WHEN WOULD YOU USE EACH MODEL?
- WHAT ARE THE MAIN ADVANTAGES OF USING CLOUD COMPUTING IN MODERN IT ENVIRONMENTS?
- COMPARE PUBLIC, PRIVATE, AND HYBRID CLOUD MODELS. WHICH ONE WOULD YOU RECOMMEND TO A BANK AND WHY?
- NAME AT LEAST THREE POPULAR CLOUD SERVICE PROVIDERS. WHAT MAKES AZURE STAND OUT IN THE DEVOPS SPACE?
- WHAT ARE THE FIVE KEY CHARACTERISTICS OF CLOUD COMPUTING? CAN YOU EXPLAIN THEM WITH SIMPLE ANALOGIES?
- IMAGINE A COMPANY WANTS TO SCALE ITS E-COMMERCE PLATFORM FOR A FESTIVAL. WHAT CLOUD AND SERVICE MODEL WOULD YOU RECOMMEND AND WHY?

DAY 2

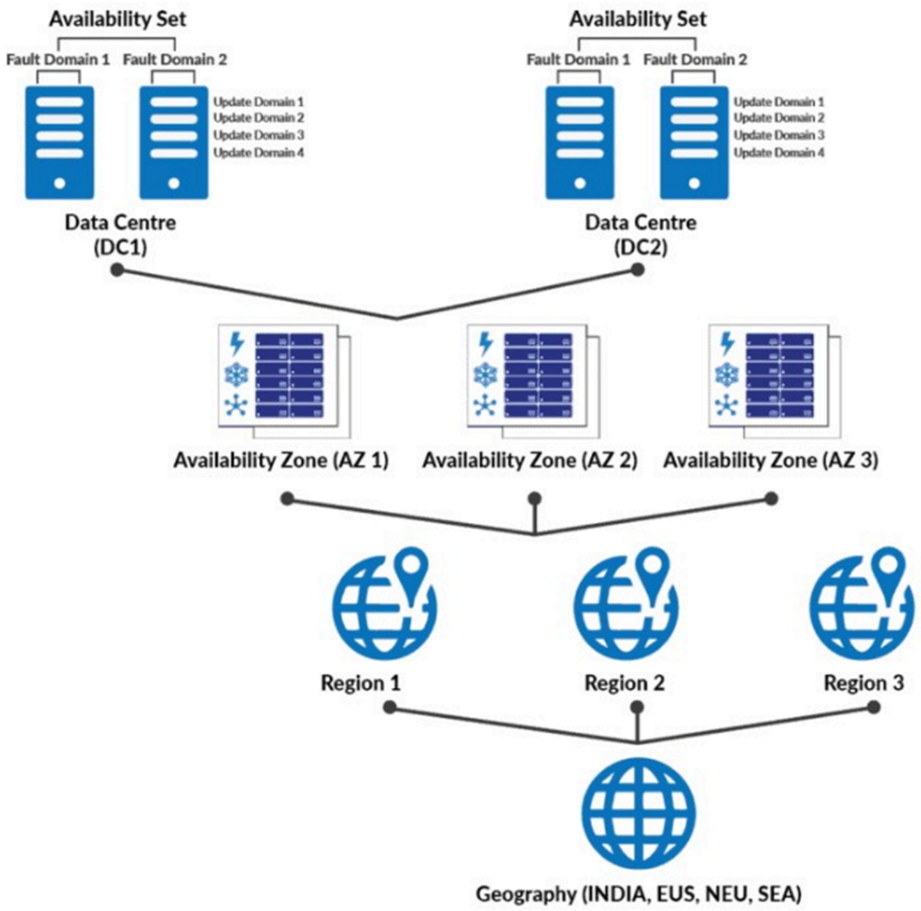

WELCOME TO
AZURE

CREATE AZURE ACCOUNT

PREREQUISITES

- GMAIL ACCOUNT OR ANY EMAIL ID
- PHONE NUMBER
- CREDIT CARD OR STUDENT MAIL ID
- INTERNET CONNECTION

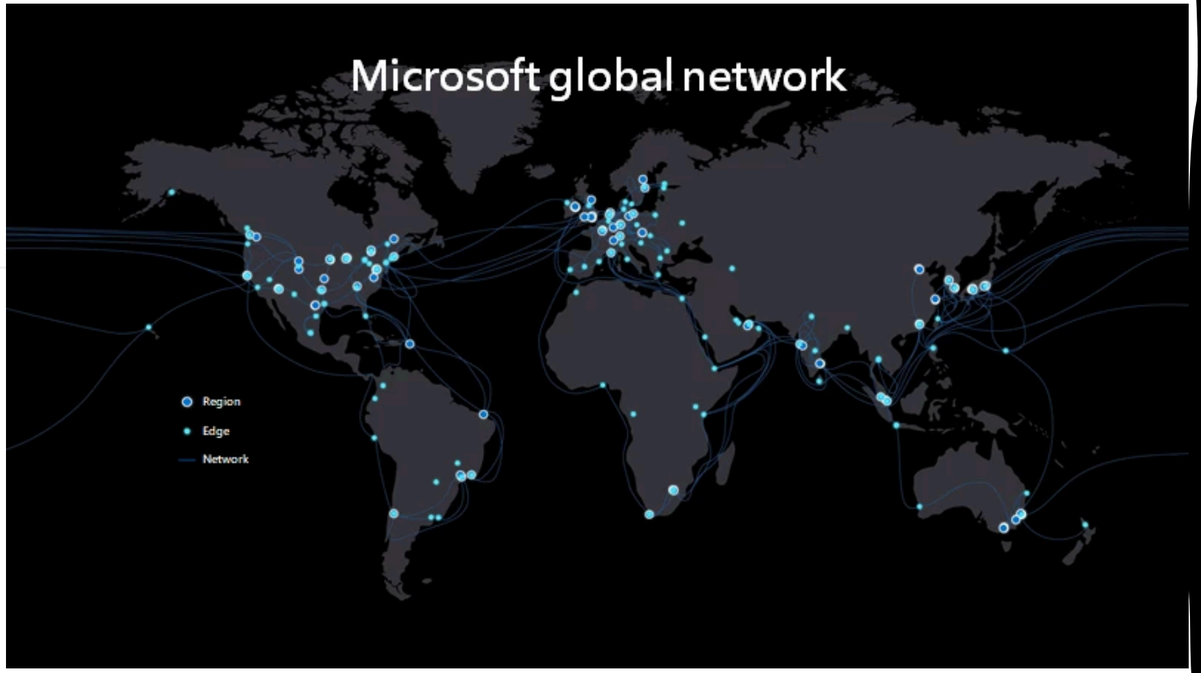
GLOBAL INFRASTRUCTURE



AZURE REGIONS

🌎 What Are Azure Regions?

- A **region** is a **set of data centers** deployed within a specific geographic location.
- Each region provides **redundant power, networking, and cooling**.
- Examples:
  - **East US**, **West Europe**, **Central India**



📌 Key Facts About Azure Regions

- Azure has **60+ regions worldwide** (more than any other cloud provider).
- Customers can choose regions **closest to users** for **low latency**.
- Each region has one or more **Availability Zones** for high availability.

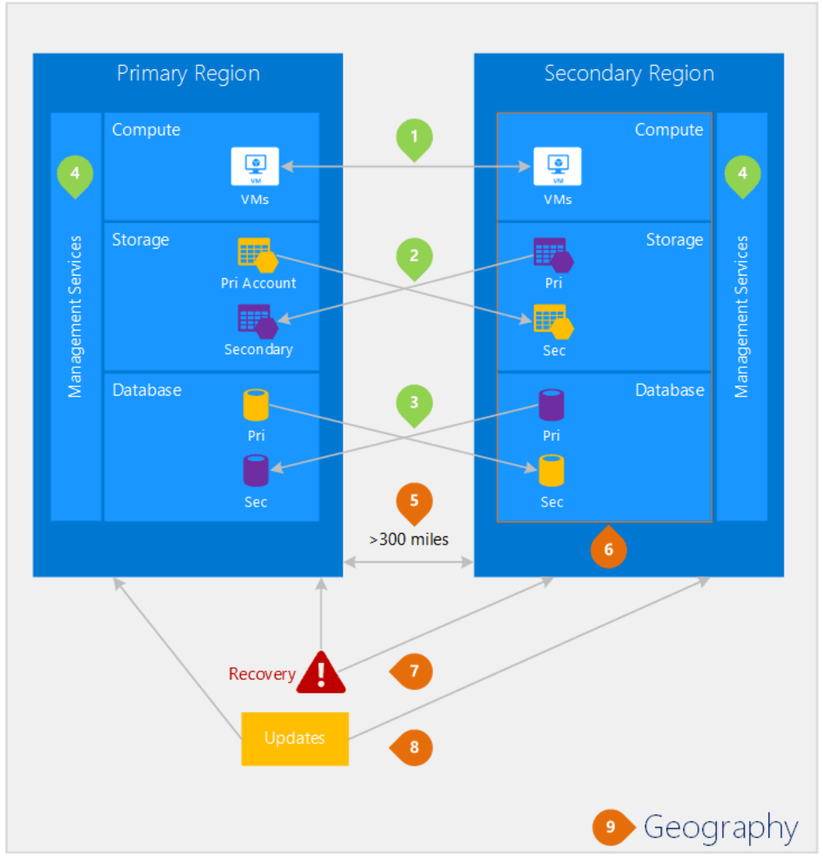
AZURE REGION PAIRS

🔗 What Are Region Pairs?

- A **region pair** is a **geographically separated pair of Azure regions.**
- Designed for **disaster recovery and data redundancy.**

✅ Benefits of Region Pairs

- One region is **prioritized for recovery** if both go down.
- **Automatic replication** for services like Azure Storage and SQL Database (Geo-redundant
- **Updates are rolled out sequentially** to minimize downtime.
- Ensures **data residency** and **compliance** (especially in government cloud).



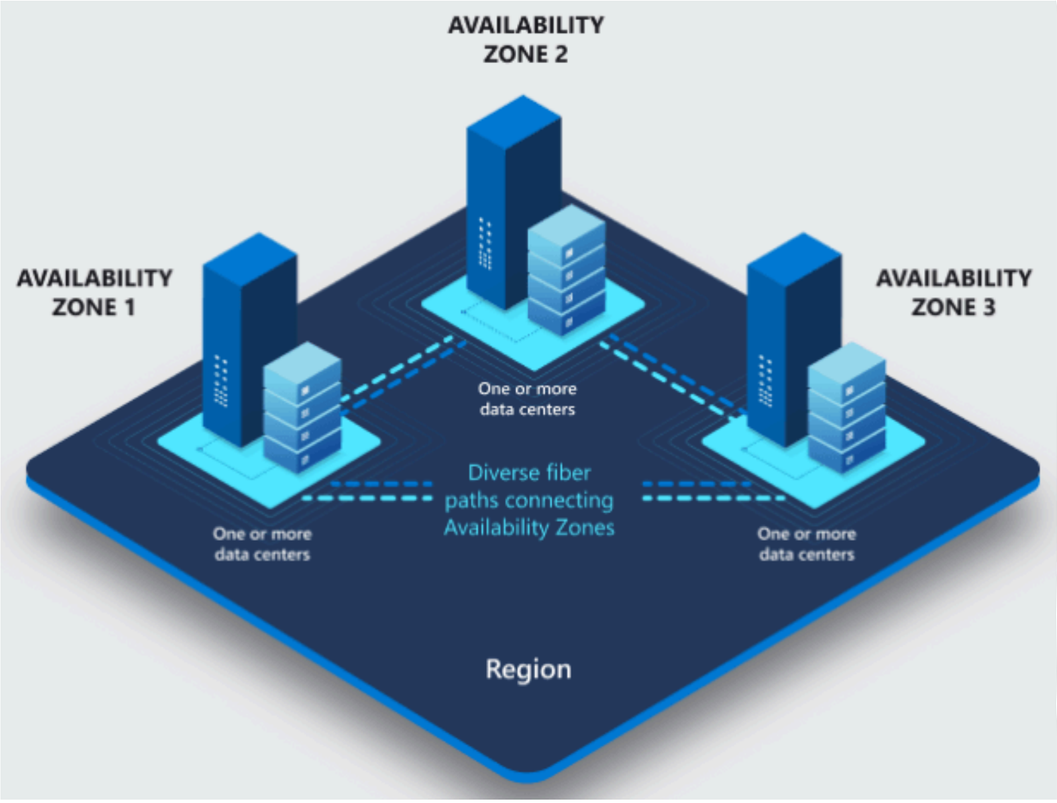
AVAILABILITY ZONES

🏢 What Are Availability Zones?

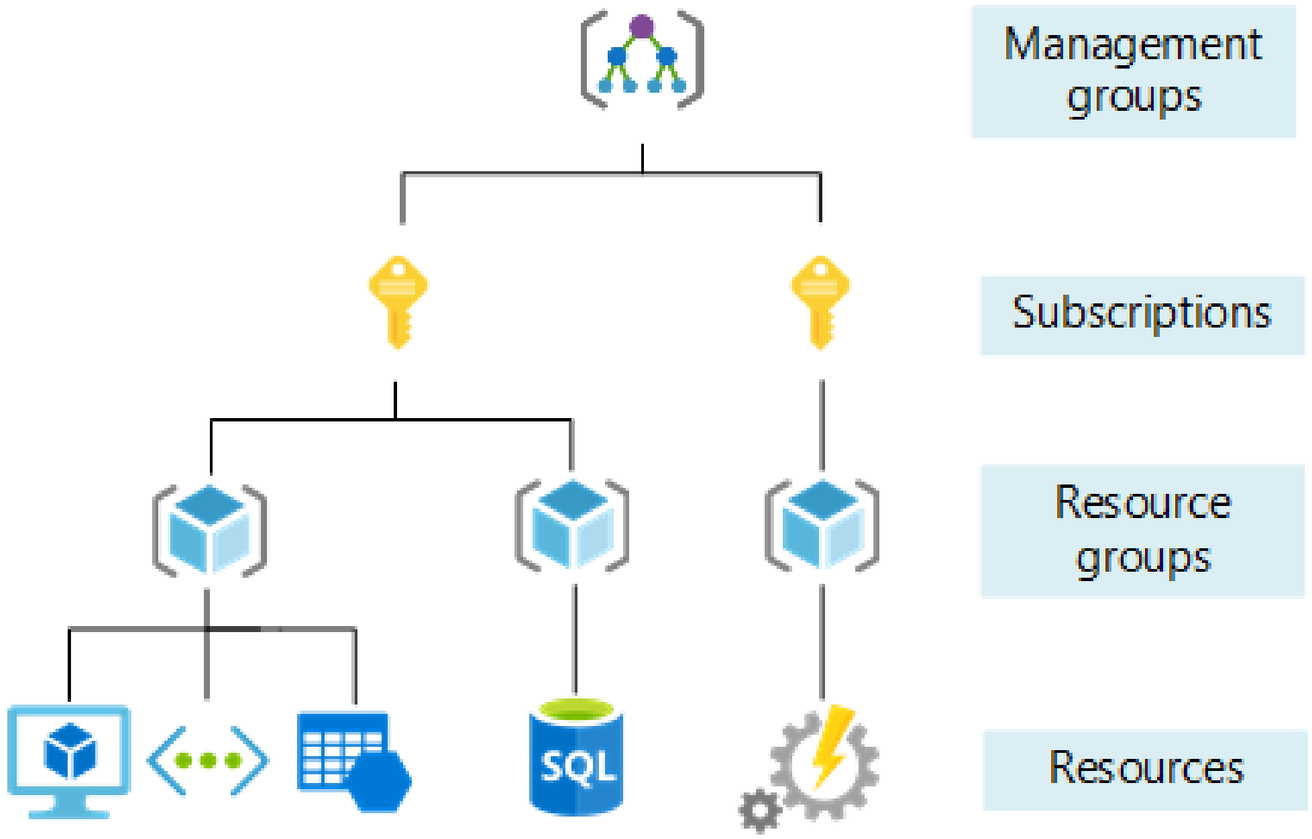
- **Physically separate data centers** within an Azure region.
- Each zone has **independent power, cooling, and networking.**
- Designed to ensure **high availability and fault tolerance.**

🧱 Structure

- Each **Azure region with AZ support** has **at least 3 zones.**
- Zones are labeled as: **Zone 1**, **Zone 2**, **Zone 3**, etc.
- They are close enough for low latency but far enough to prevent failure impact.



AZURE ACCOUNT HIREARCHY



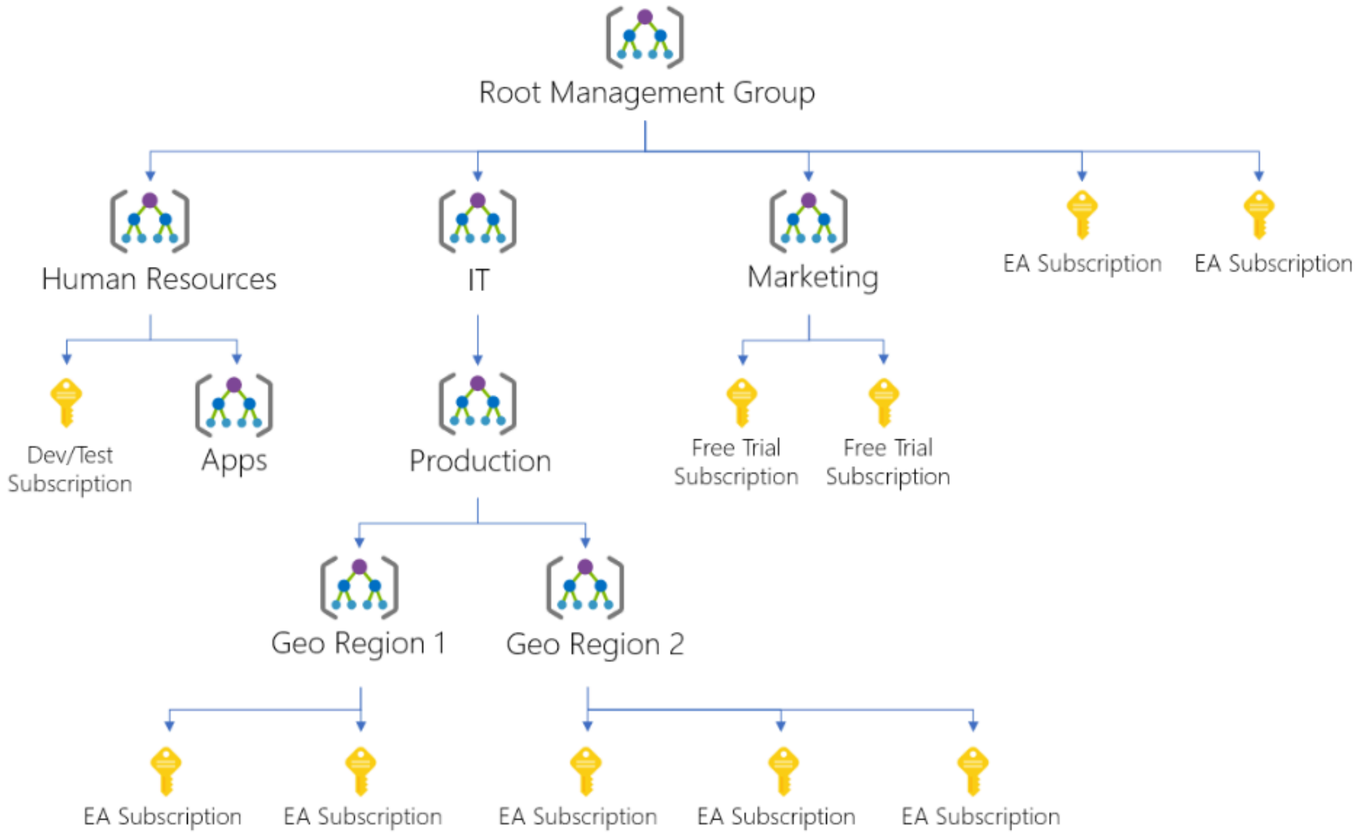
Management groups

Subscriptions

Resource groups

Resources

MANAGEMENT GROUPS



CREATE AZURE ACCOUNT

LAB TOPICS

- VNET
- SUBNETTING
- NSG'S
- CREATING A LINUX VM(UBUNTU SERVER)
- HOST A STATIC WEBSITE

INTERVIEW QUESTIONS

WHAT IS AN AZURE SUBSCRIPTION?
CAN A SINGLE AZURE ACCOUNT HAVE MULTIPLE SUBSCRIPTIONS? WHY WOULD YOU DO THIS?
HOW DO YOU MANAGE ACCESS WITHIN A SUBSCRIPTION?
WHAT ARE AZURE MANAGEMENT GROUPS?
HOW DEEP CAN THE MANAGEMENT GROUP HIERARCHY GO?
HOW IS ACCESS INHERITED IN MANAGEMENT GROUPS?
WHAT IS A VIRTUAL NETWORK (VNET) IN AZURE?
WHAT IS THE PURPOSE OF A SUBNET IN A VNET?
CAN TWO SUBNETS IN THE SAME VNET COMMUNICATE WITH EACH OTHER?
WHAT IS AN NSG IN AZURE?
WHERE CAN YOU APPLY AN NSG?
WHAT HAPPENS IF AN NSG IS APPLIED AT BOTH SUBNET AND NIC LEVELS?
CAN YOU DENY SPECIFIC IP RANGES USING NSGS?
HOW WOULD YOU ISOLATE A GROUP OF VMS IN A SUBNET FROM INTERNET ACCESS?

BASTION HOST

BASTION HOST ON AZURE IS A FULLY MANAGED PAAS SERVICE THAT ALLOWS YOU TO SECURELY ACCESS YOUR VIRTUAL MACHINES (VMS) USING REMOTE DESKTOP PROTOCOL (RDP) OR SECURE SHELL (SSH) DIRECTLY THROUGH THE AZURE PORTAL, WITHOUT EXPOSING PUBLIC IP ADDRESSES.

WITH AZURE BASTION, YOU CAN:

- ELIMINATE THE NEED FOR PUBLIC IPS ON VMS.
- AVOID OPENING INBOUND PORTS ON NSGS.
- REDUCE EXPOSURE TO THREATS

NAT GATEWAY

A NAT GATEWAY IN AZURE IS A FULLY MANAGED NETWORK ADDRESS TRANSLATION (NAT) SERVICE THAT ENABLES OUTBOUND INTERNET CONNECTIVITY FOR VIRTUAL MACHINES (VMS) OR OTHER COMPUTE RESOURCES WITHOUT REQUIRING PUBLIC IP ADDRESSES ON EACH RESOURCE.

AZURE TRANSLATES THE PRIVATE IP TO ONE OF THE PUBLIC IPS ASSIGNED TO THE NAT GATEWAY.

VPC PEERING

VNET PEERING IS A NETWORKING FEATURE IN AZURE THAT LETS YOU CONNECT TWO OR MORE VIRTUAL NETWORKS (VNETS) PRIVATELY THROUGH AZURE'S BACKBONE NETWORK. THIS ALLOWS RESOURCES IN DIFFERENT VNETS TO COMMUNICATE WITH EACH OTHER USING PRIVATE IP ADDRESSES, JUST LIKE THEY WOULD WITHIN THE SAME VNET.

ONCE TWO VNETS ARE PEERED:

- THEY APPEAR AS ONE NETWORK FOR ROUTING PURPOSES.
- TRAFFIC BETWEEN THEM IS PRIVATE AND ENCRYPTED.
- COMMUNICATION IS LOW-LATENCY AND HIGH-BANDWIDTH, VIA AZURE'S INTERNAL INFRASTRUCTURE

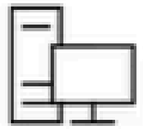
INTERVIEW QUESTIONS

- WHAT IS A VIRTUAL NETWORK (VNET) IN AZURE AND HOW DOES IT WORK?
- WHAT ARE THE KEY DIFFERENCES BETWEEN A VNET AND A TRADITIONAL ON-PREMISE NETWORK?
- HOW DO YOU DESIGN SUBNETTING WITHIN A VNET AND WHY IS IT IMPORTANT?
- WHAT ARE THE SUPPORTED IP ADDRESS RANGES FOR VNETS?
- HOW IS ROUTING HANDLED BETWEEN SUBNETS IN THE SAME VNET?
- WHAT IS A NETWORK SECURITY GROUP (NSG) IN AZURE?
- WHAT IS THE DIFFERENCE BETWEEN NSG RULES AND FIREWALL RULES?
- WHAT ARE THE DEFAULT NSG RULES, AND CAN YOU OVERRIDE THEM?
- HOW DO NSGS EVALUATE TRAFFIC (PRIORITY, DIRECTION, AND RULES)?
- HOW CAN YOU APPLY NSGS – AT THE NIC LEVEL VS SUBNET LEVEL? WHAT'S THE IMPACT?
- WHAT IS AN APPLICATION SECURITY GROUP (ASG)?
- HOW DO ASGS SIMPLIFY NETWORK SECURITY MANAGEMENT?
- CAN ASGS SPAN MULTIPLE VNETS? WHY OR WHY NOT?
- HOW ARE ASGS USED IN COMBINATION WITH NSGS?
- WHAT IS VNET PEERING AND HOW IS IT DIFFERENT FROM VPN GATEWAY?
- WHAT ARE THE KEY BENEFITS OF VNET PEERING?
- WHAT IS GLOBAL VNET PEERING, AND HOW DOES IT WORK?
- WHAT ARE THE LIMITATIONS OF VNET PEERING?
- CAN PEERED VNETS HAVE OVERLAPPING IP RANGES? WHY OR WHY NOT?

SLA's

VM SLA

99.9% with Premium Storage

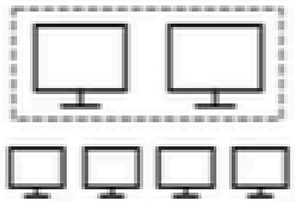


SINGLE VM

Easier lift and shift

VM SLA

99.95%

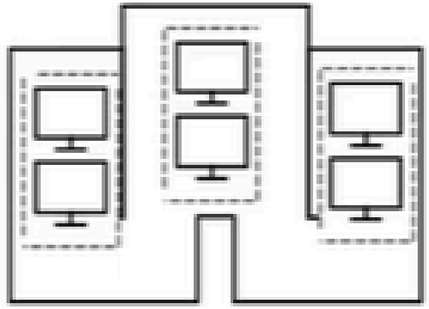


AVAILABILITY SETS

Protecting against failures within datacenters

VM SLA

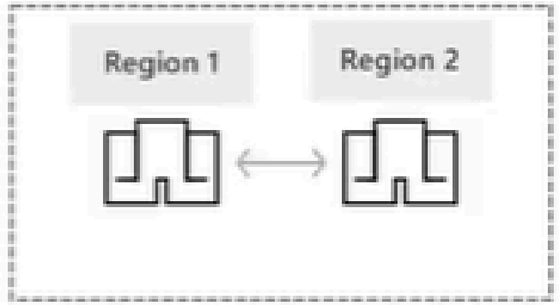
99.99%



AVAILABILITY ZONES

Protection from entire datacenter failures

MULTI-REGION DISASTER RECOVERY



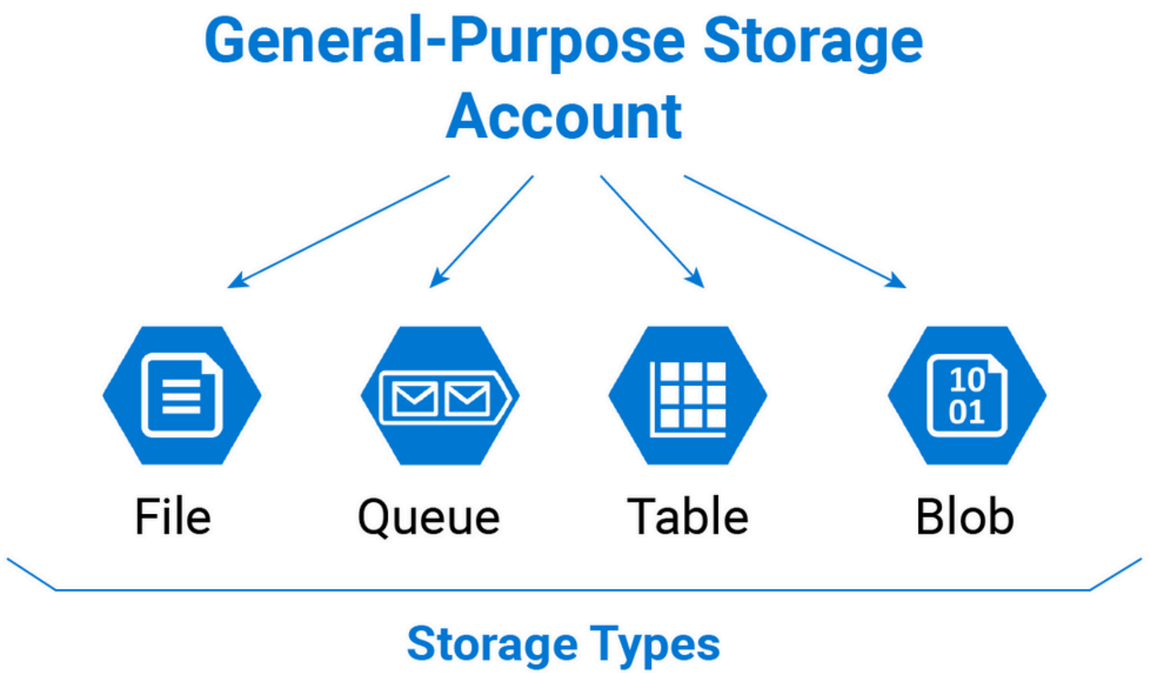
REGION PAIRS

Regional protection within Data Residency Boundaries

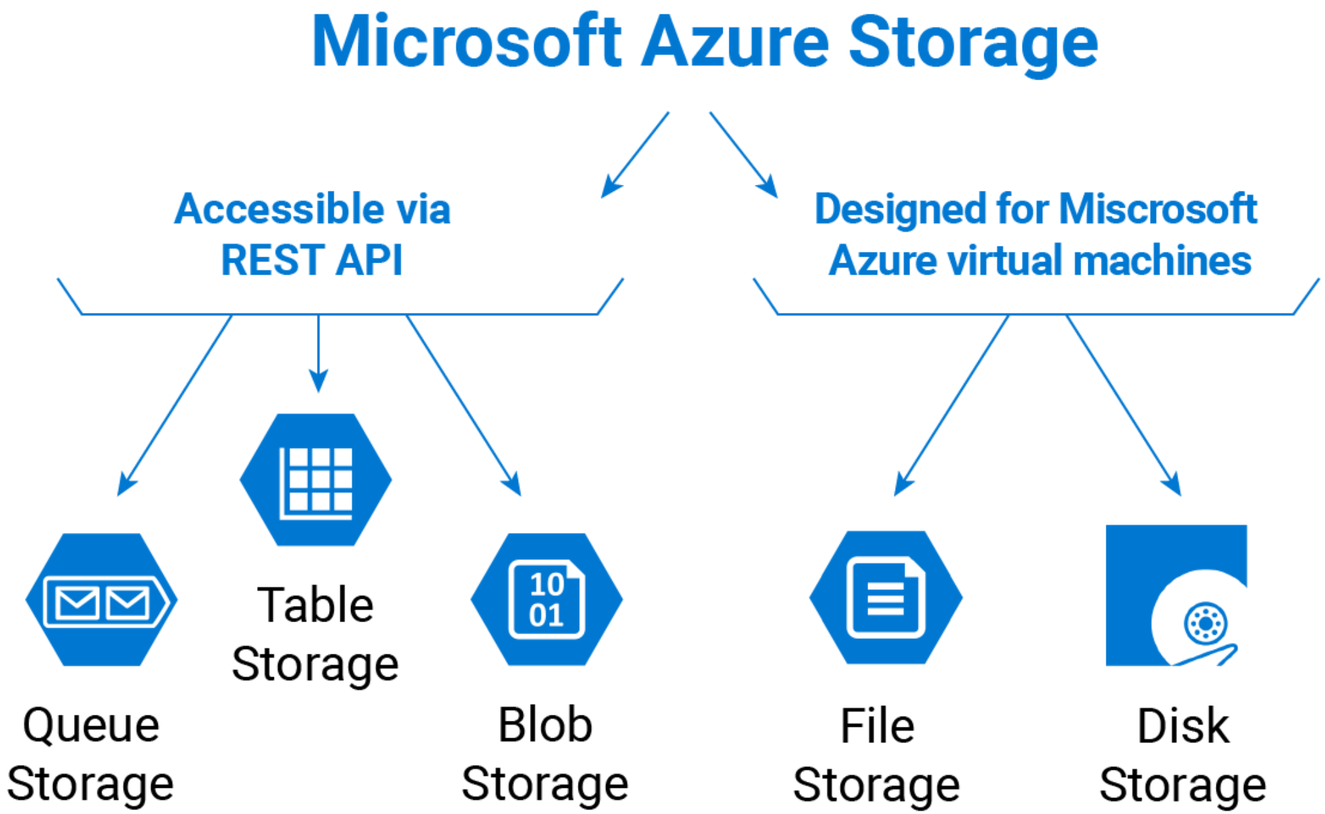
INTERVIEW QUESTIONS-VM

- WHAT IS A ARE THE KEY DIFFERENCES BETWEEN AVAILABILITY SETS AND AVAILABILITY ZONES IN AZURE?
- HOW DOES AZURE ENSURE HIGH AVAILABILITY WHEN DEPLOYING VMS ACROSS DIFFERENT FAULT DOMAINS AND UPDATE DOMAINS?
- YOU ARE ASKED TO DEPLOY A HIGHLY AVAILABLE APPLICATION – WOULD YOU USE A SCALE SET OR AVAILABILITY SET? WHY?
- HOW DO YOU CONFIGURE AN AZURE VM TO BE ACCESSIBLE ONLY THROUGH A SECURE METHOD WITHOUT ASSIGNING A PUBLIC IP?
- WHAT IS A VIRTUAL MACHINE SCALE SET (VMSS) AND HOW DOES IT HELP WITH AUTOSCALING?
- CAN AN EXISTING VM BE MOVED INTO AN AVAILABILITY SET OR AVAILABILITY ZONE AFTER CREATION? EXPLAIN YOUR ANSWER.
- DESCRIBE THE PROCESS OF CREATING A LINUX VM IN A SPECIFIC AVAILABILITY ZONE USING THE AZURE PORTAL.
- WHAT ARE THE USE CASES WHERE VMSS (VIRTUAL MACHINE SCALE SETS) ARE PREFERRED OVER INDIVIDUAL VMS?
- EXPLAIN HOW AZURE LOAD BALANCER WORKS WITH VM SCALE SETS.
- HOW DOES AZURE BASTION IMPROVE THE SECURITY POSTURE OF YOUR VM CONNECTIVITY?
- YOU DEPLOYED 3 VMS IN AN AVAILABILITY SET. WHAT WOULD BE THE MAXIMUM NUMBER OF UPDATE DOMAINS AND FAULT DOMAINS USED?
- HOW DO YOU CONNECT TO A LINUX VM HOSTED IN AZURE WITHOUT EXPOSING PORT 22 TO THE INTERNET?
- WHAT HAPPENS IF AN ENTIRE AVAILABILITY ZONE GOES DOWN? HOW CAN YOU DESIGN YOUR VM DEPLOYMENT TO BE RESILIENT?
- WHAT IS THE DEFAULT NUMBER OF FAULT DOMAINS AND UPDATE DOMAINS IN AN AVAILABILITY SET, AND CAN THEY BE CHANGED?
- HOW WOULD YOU AUTOMATE THE DEPLOYMENT OF 5 VMS IN DIFFERENT AVAILABILITY ZONES FOR DISASTER RECOVERY PURPOSES?
- WHAT ARE SOME OF THE LIMITATIONS OF USING VIRTUAL MACHINE SCALE SETS?
- HOW DO YOU APPLY CUSTOM SCRIPTS DURING VM CREATION TO CONFIGURE THE OS OR INSTALL SOFTWARE?
- IF YOU OBSERVE THAT VM CONNECTIONS ARE INTERMITTENTLY FAILING DURING UPDATES, WHAT MIGHT BE THE CAUSE?
- EXPLAIN THE DIFFERENCE BETWEEN ‘UNIFORM’ AND ‘FLEXIBLE’ ORCHESTRATION MODES IN VM SCALE SETS.
- WHAT ARE THE KEY METRICS OR CONDITIONS YOU CAN USE TO CONFIGURE AUTO-SCALING RULES IN A VM SCALE SET?

STORAGE ACCOUNTS



STORAGE ACCOUNTS



Storage Account



Table

- Ideal for structured and non-relational data
- **URL - http://<storage account>.table.core.windows.net/<table>**



Blob

- Ideal storing massive amounts of unstructured data, such as text or binary data
- **URL - http://<storage account>.blob.core.windows.net**



Queue

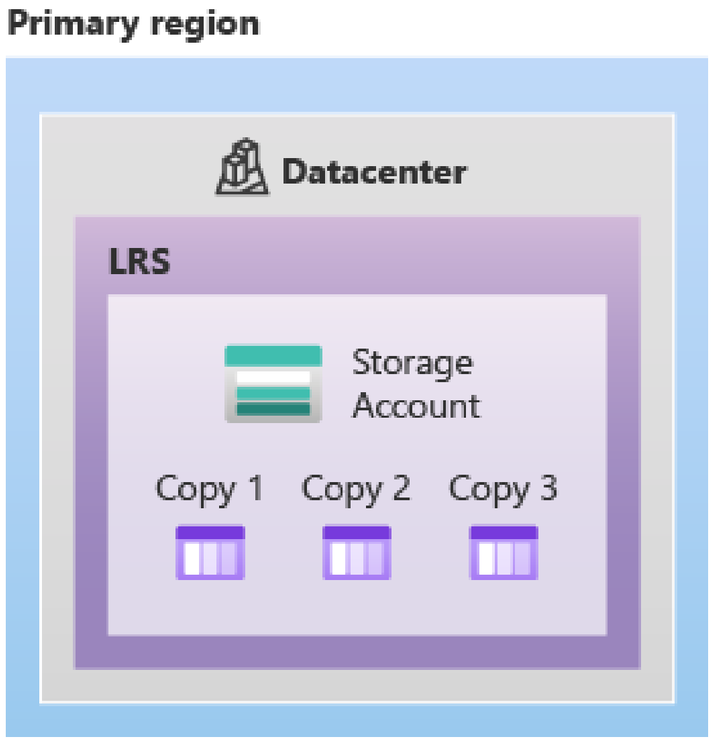
- Ideal for reliable messaging for communication between applications or application components
- **URL - http://<storage account>.queue.core.windows.net/<queue>**

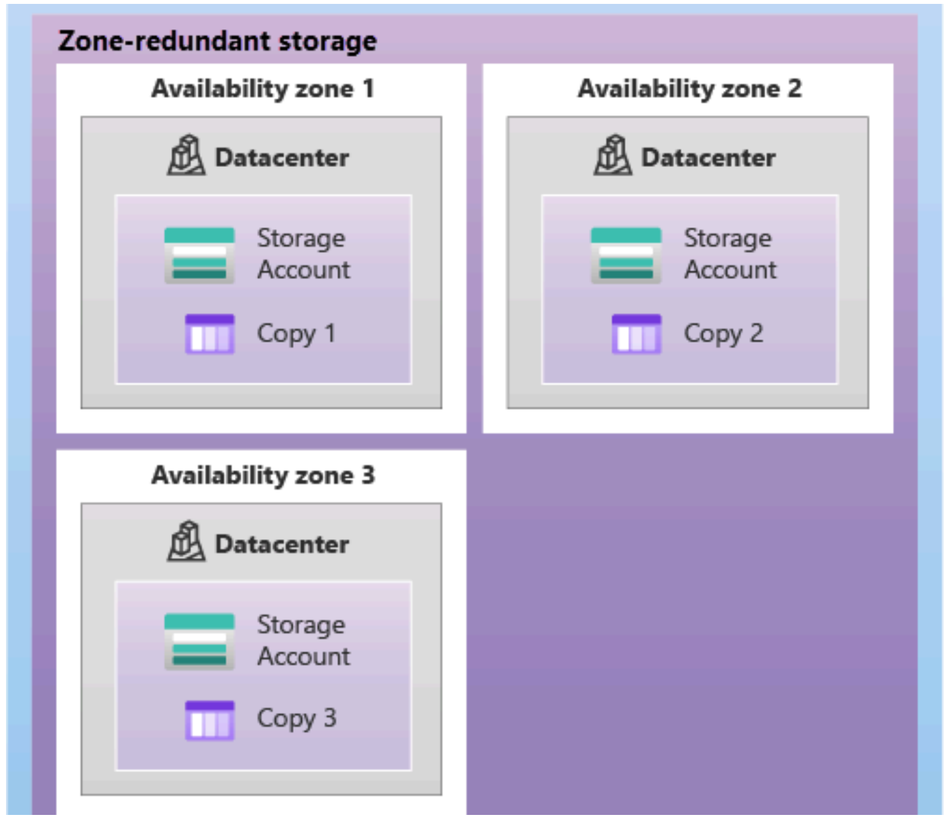


Files

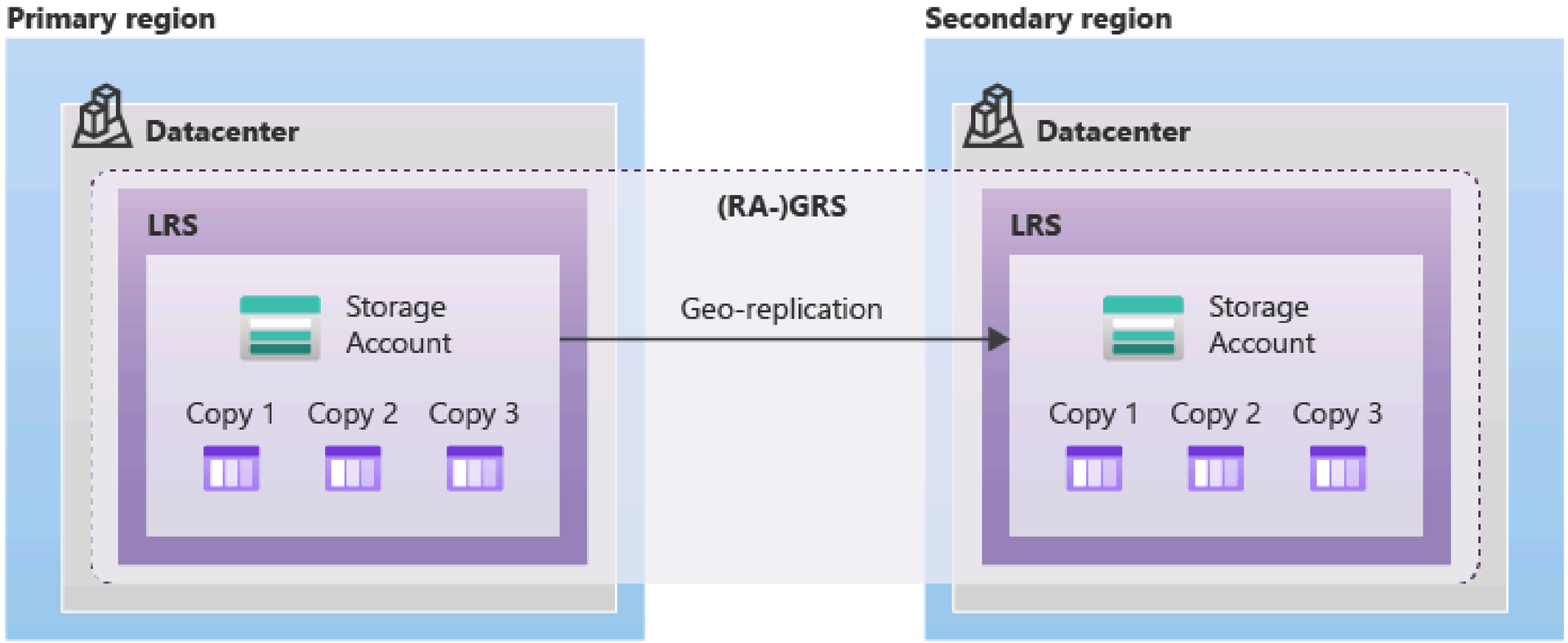
- A Simple, secure and fully managed file share in the cloud accessible via SMB
- **URL - https://<storage-account>.file.core.windows.net/<file-share>**

STORAGE ACCOUNTS

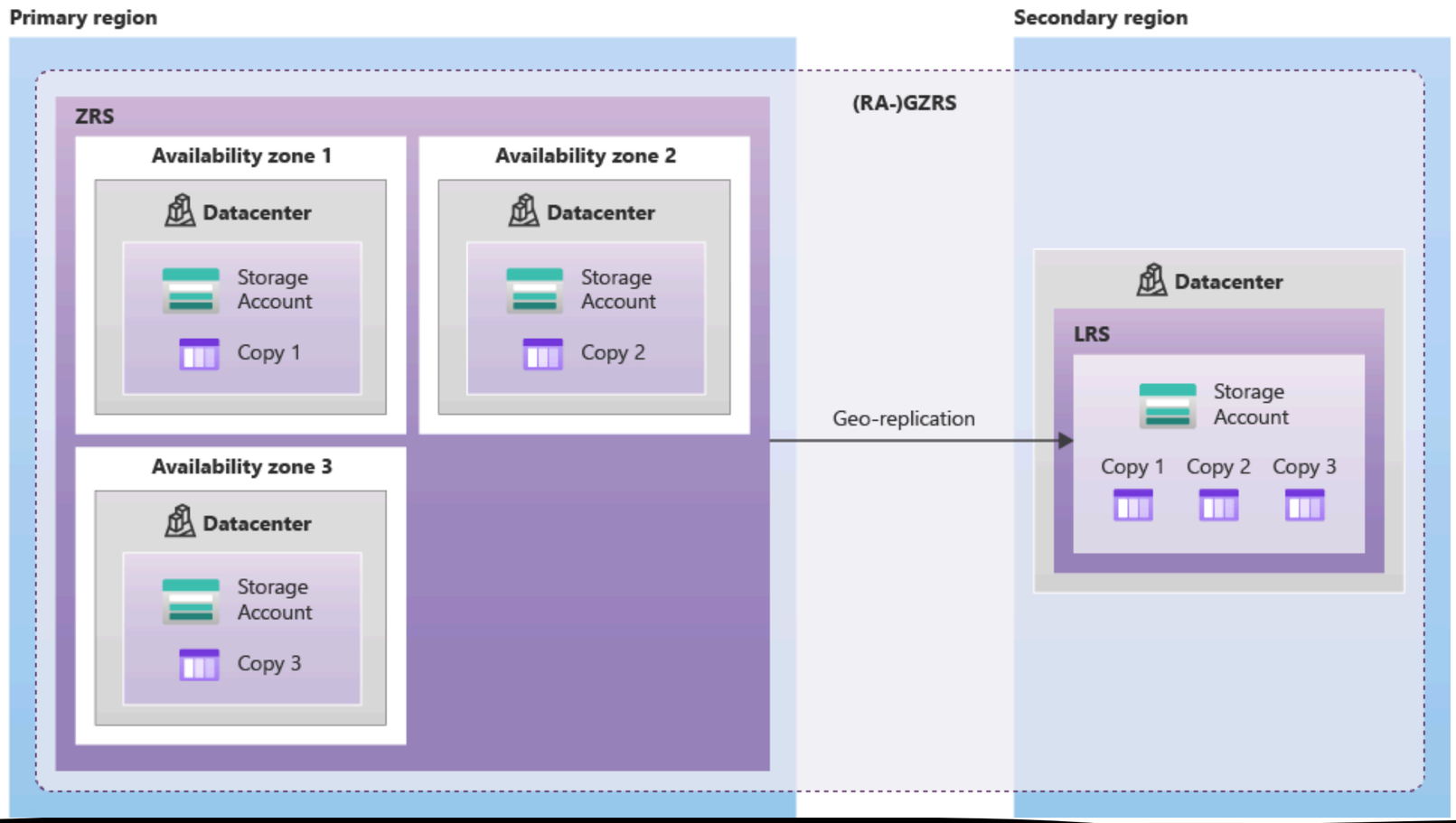




STORAGE ACCOUNTS



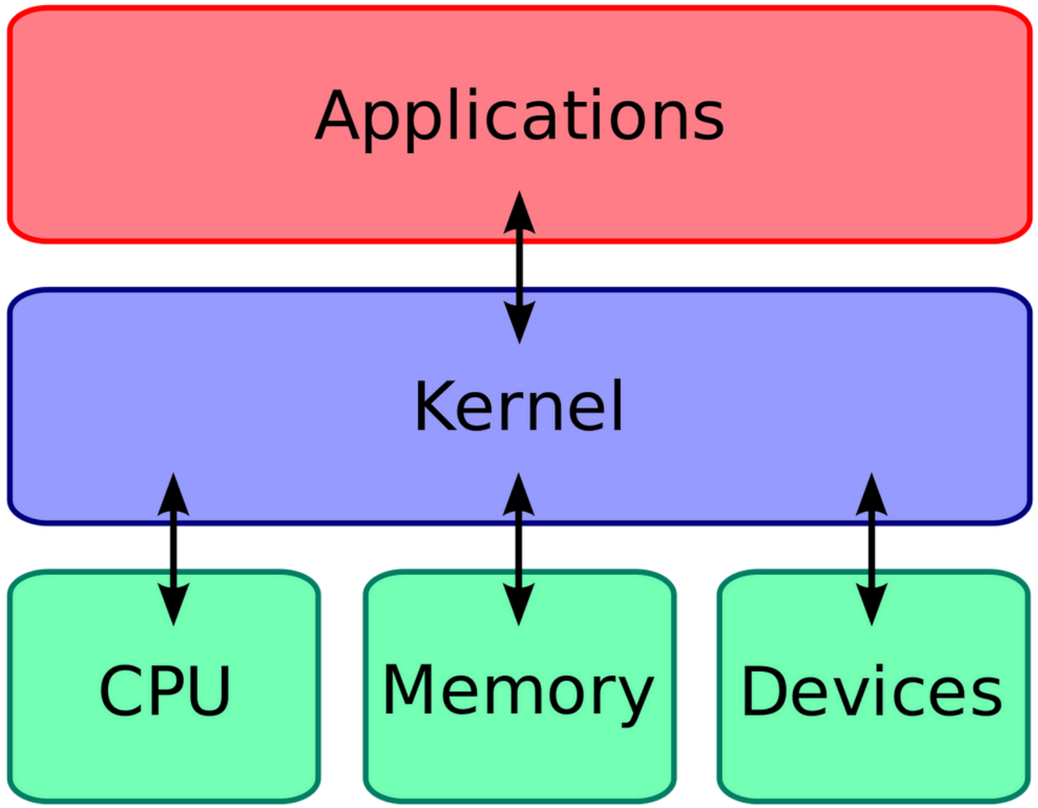
STORAGE ACCOUNTS




LINUX

LINUX KERNEL

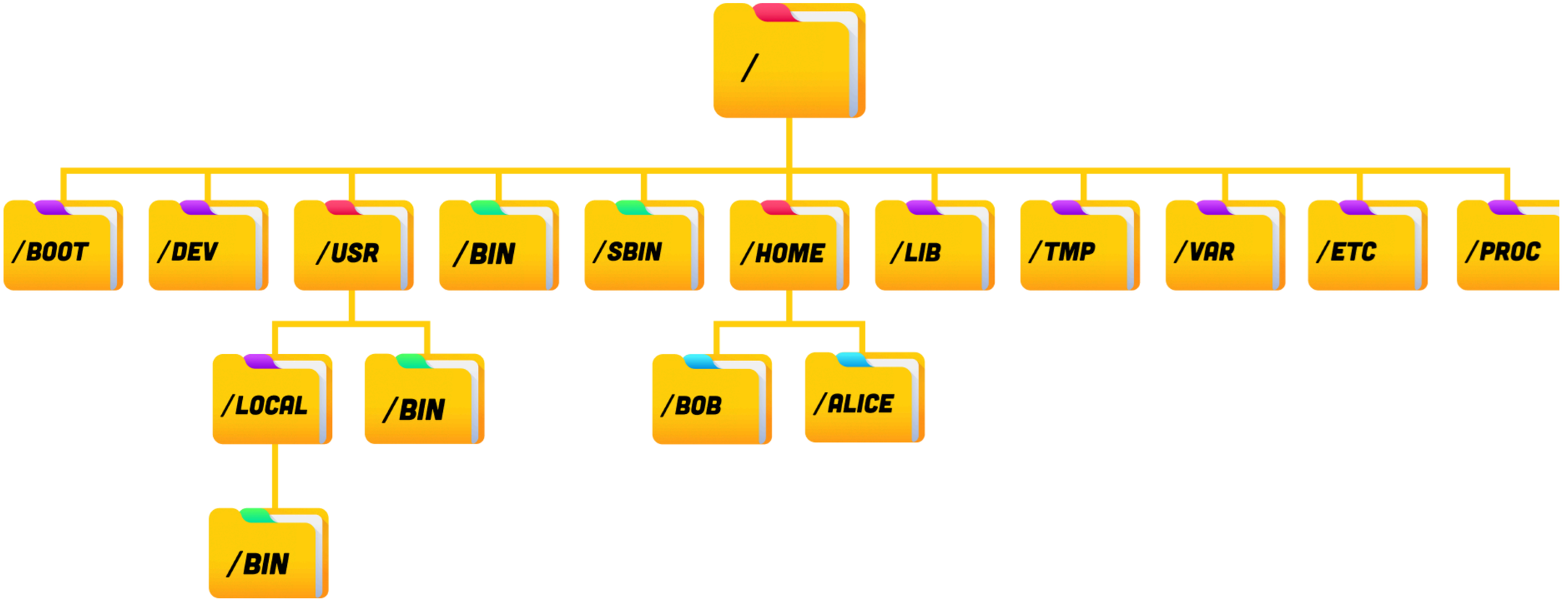
- Memory Management
- Process Management
- Drivers
- Processes



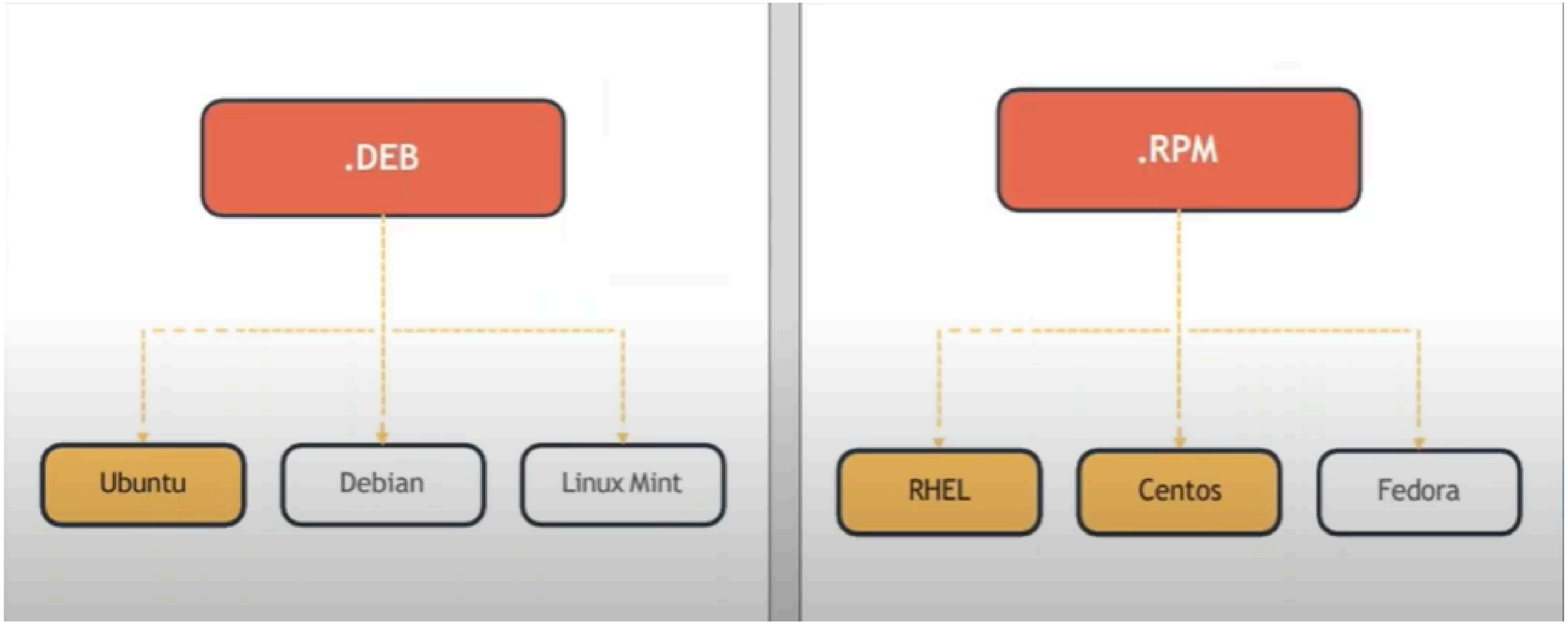
LINUX VERSIONS



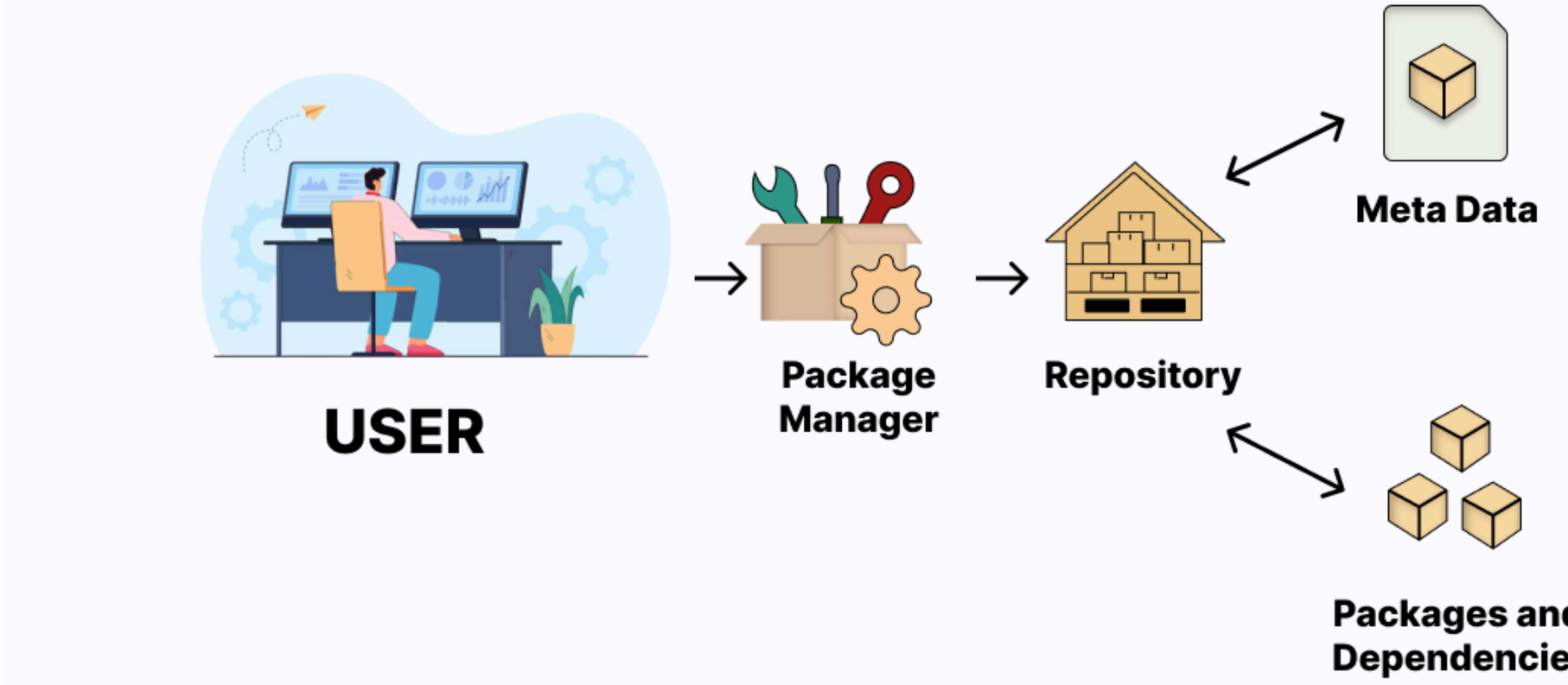
LINUX DIRECTORIES



LINUX PACKAGES



LINUX PACKAGES



APT AND YUM

Tool	Stands For	Used By
`apt`	Advanced Package Tool	Ubuntu, Debian
`yum`	Yellowdog Updater Modified	CentOS, RHEL

- HANDLES DEPENDENCIES AUTOMATICALLY
- EASIER TO SEARCH, INSTALL, UPGRADE
- SAFER: PACKAGE FROM VERIFIED REPOSITORIES
- MAINTAINS A LOCAL CACHE OF METADATA
- SUPPORTS INTERACTIVE OR AUTOMATED INSTALLS


GIT

VERSION CONTROL

- VERSION CONTROL, ALSO KNOWN AS SOURCE CONTROL, IS THE PRACTICE OF TRACKING AND MANAGING CHANGES TO SOFTWARE CODE.
- VERSION CONTROL SOFTWARE KEEPS TRACK OF EVERY MODIFICATION TO THE CODE IN A SPECIAL KIND OF DATABASE.
- VERSION CONTROL TRACKS EVERY DEVELOPER CHANGE AND HELPS PREVENT CONCURRENT WORK FROM CONFLICTING.

BENEFITS OF THE VERSION CONTROL SYSTEM

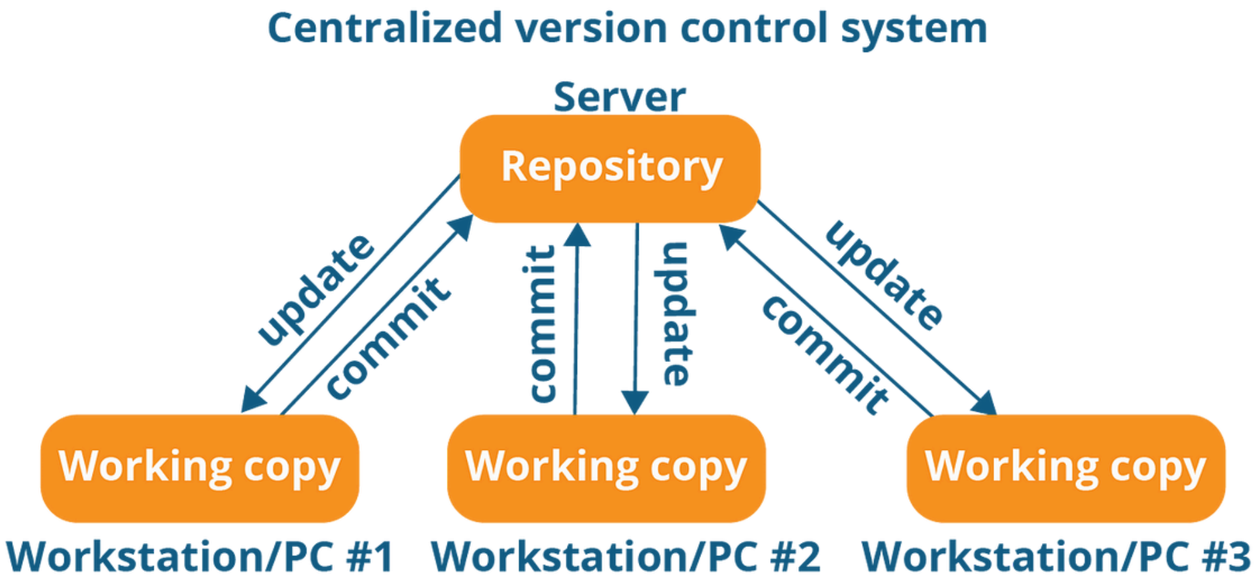
- Enhances the project development speed by providing efficient collaboration.
- Reduces possibilities of errors and conflicts meanwhile project development through traceability to every small change.
- Employees or contributors of the project can contribute from anywhere irrespective of the different geographical locations through this VCS.
- For each different contributor to the project, a different working copy is maintained and not merged to the main file unless the working copy is validated. The most popular example is Git, Helix Core, Microsoft TFS.
- Helps in recovery in case of any disaster or contingent situation.
- Informs us about Who, What, When, and Why changes have been made.

CENTRALIZED VCS

- Centralized version control systems contain just one repository globally, and every user needs to commit to reflect one's changes in the repository. It is possible for others to see your changes by updating.
- Two things are required to make your changes visible to others, which are:

☑ You commit

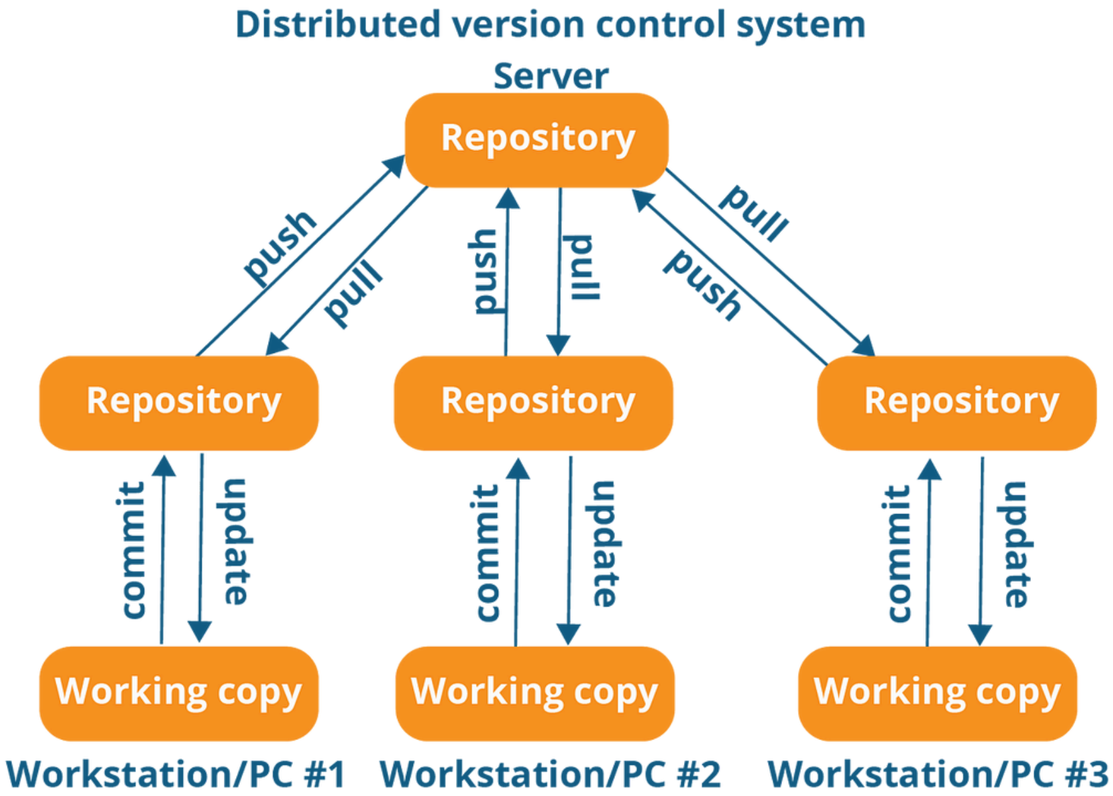
☑ They update



CENTRALIZED VCS

- **Distributed version control systems contain multiple repositories.**
- **Each user has their repository and working copy. Users need to pull to fetch the latest changes, whereas developers need to push their changes to update the remote repository.**

- ☑ **commit**
- ☑ **push**
- ☑ **pull**
- ☑ **update**



BENEFITS OF GIT OVER SVN

- GIT utilizes multiple repositories: a central repository and a series of local repositories.
- Local repositories are exact copies of the central repository, complete with the entire history of changes, unlike SVN.
- The Git workflow is similar to SVN, but with an extra step:
- To create a new feature, you take an exact copy of the central repository to create your local repository on your local machine.

Why many developers prefer GIT:

- It's faster to commit
- No more single point of failure
- If the centralized server is down, developers can still work using a local copy
- It's available offline

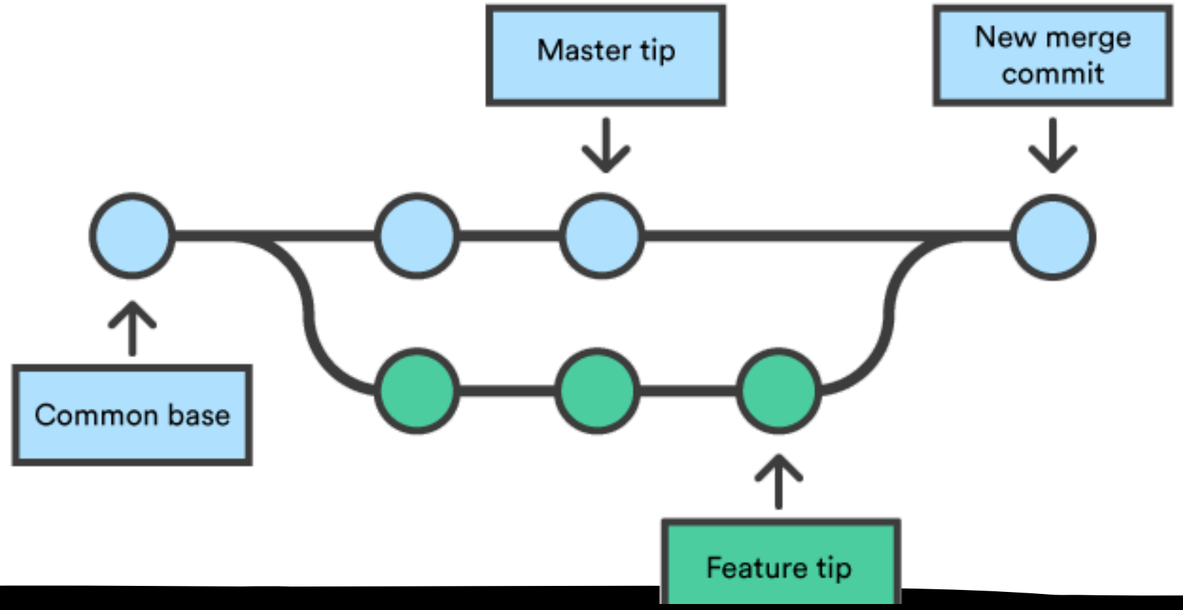
- In SVN, the total number of branches = total number of physical folders
- In GIT, the total number of branches is logical — only the reference or pointer moves ahead or back with each branch

GIT COMMANDS

- git init
- git config
- git add
- git commit
- git status
- gitk (Graphical representation)
- git log
- git show
- git branch
- git switch
- git clone
- git checkout
- git merge / git rebase
- git pull / git push / git fetch
- git remote
- git cherry-pick
- git reset
- git stash
- git tag
- git diff

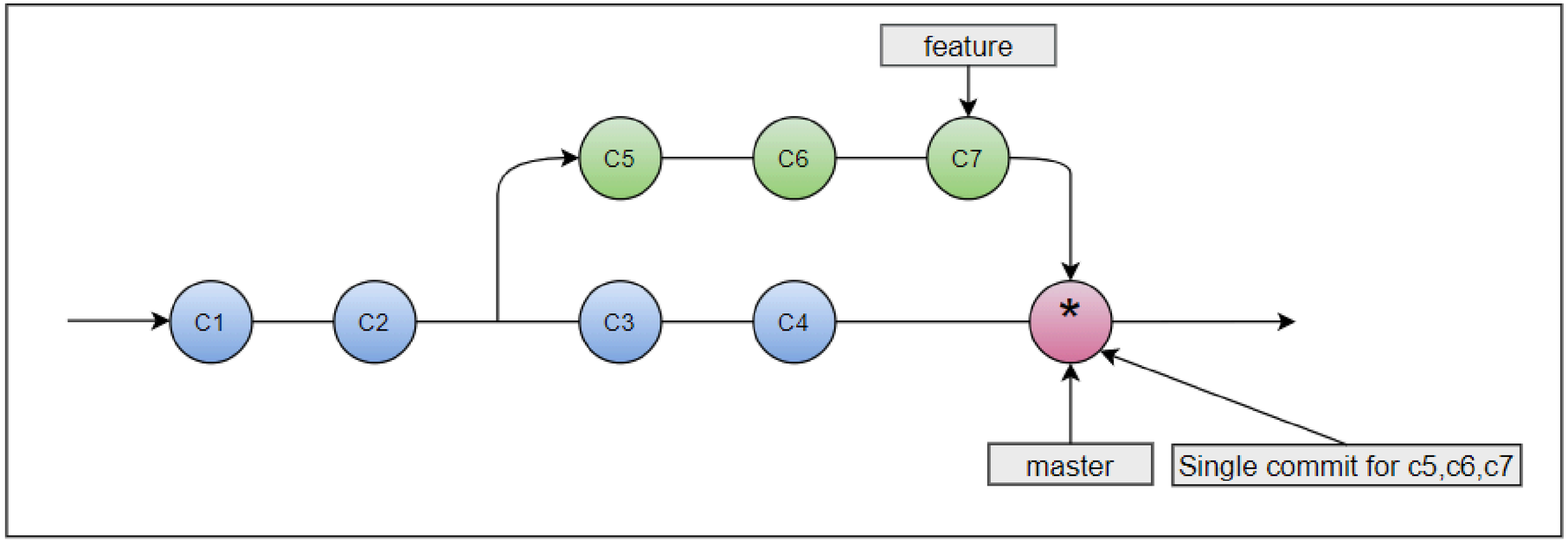
MERGE (NO FAST-FORWARD):

- It is a default integration strategy in Azure Repos, GitHub, and most other Git providers
- All individual commits are preserved as-is, and a new merge commit is created
- This strategy is helpful because it illustrates exactly how a developer (or developers) worked on a pull request, including each individual commit along the way



SQUASH COMMIT

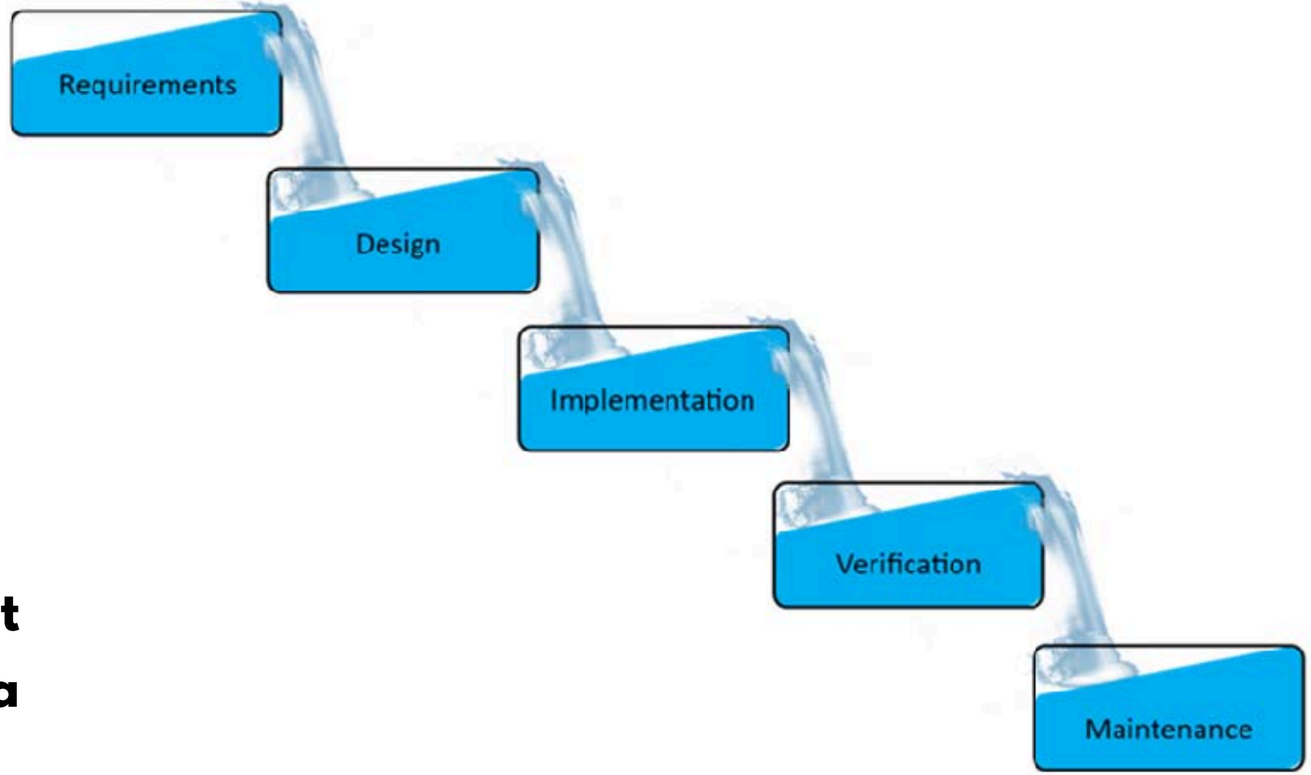
- In Squash, each pull request becomes a single commit in master, and there are no merges — just a simple, straight, linear history
- Individual commits are lost, which is best for teams that use “fix-up” commits or do not carefully craft individual commits for review before pushing them




WORK ITEMS

SOFTWARE DEVELOPMENT

- Software is built in stages — from gathering requirements, designing, coding, testing, deploying, and finally maintaining it in production. This complete flow is known as the Software Development Life Cycle (SDLC)
- The Waterfall model is a step-by-step, linear approach where one phase must complete before the next starts. Once a phase is done, there is no going back.



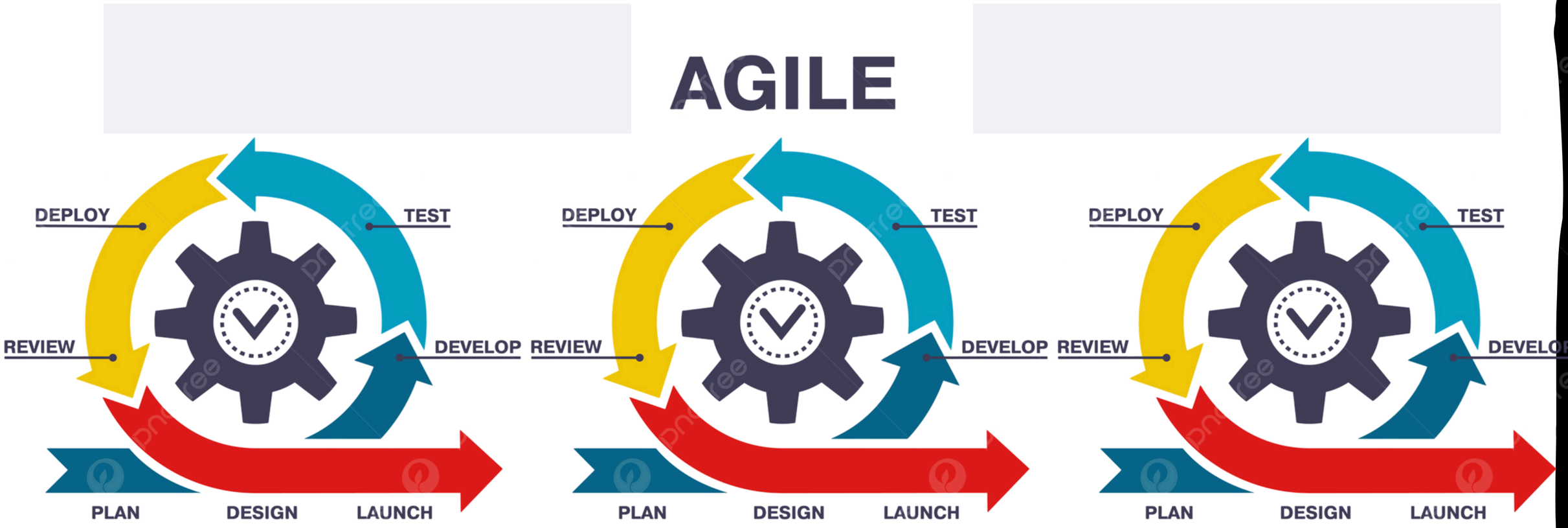
Problems with the Waterfall Model

- Customer feedback comes only at the end.
- Changes are hard to implement.
- Late testing leads to bugs being discovered very late.
- Long delivery cycles.

Agile Came into Picture

- To solve Waterfall's issues, Agile was introduced. It focuses on speed, flexibility, and continuous feedback. Work is done in small cycles, called iterations or sprints.
- Scrum is a type of Agile framework. Work is divided into 2-4 week sprints, with daily meetings, planning, and regular customer feedback.
- **Difference Between Scrum and Agile**
  - Agile is a mindset or philosophy.
  - Scrum is a structured framework that follows Agile principles.
  - Think of Agile as a set of values and Scrum as a way to follow those values practically.

Agile Framework



Why Scrum is Fast

- An iteration (also called a sprint in Scrum) is a short cycle where a small set of features is planned, developed, tested, and reviewed.

- Because of:
  - Short sprints
  - Regular updates
  - Team collaboration
  - Quick response to feedback

Scrum Process



Challenges Before DevOps

- Developers are fast; infrastructure setup is slow.
- Code movement from Dev → QA → Production is delayed.
- Application misbehavior post-deployment.
- Developers constantly pulled in for debugging.
- Lack of monitoring.
- Low-quality delivery due to last-minute changes.

DevOps to Rescue

Dev and Infra teams work together from the start so that infra is ready before development ends.

- Collaboration
- Automation
- Shared responsibility
- Continuous improvement

Role of DevOps Engineer

- Set up CI/CD pipelines
- Automate infrastructure
- Implement monitoring/logging
- Troubleshoot deployment issues
- Work closely with Dev and Ops